Task 1: Geocoding Office Data

Whole US Implementation

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1 This Week's Progress

This report is a follow-up to the 03/30 report and addresses a bug in the previous match-statistics summary. In that earlier report, the "one-to-one" category was misleading: it was presented as if it represented successful matches, when in fact it conflated three distinct outcomes: matched, did-not-match, and missing rows. This report corrects that classification and presents the updated numbers, with each outcome reported separately.

This report also updates the methodology of the matching algorithm: rather than returning the full geocoder match record, the algorithm is now scoped to output only the resolved census tract for each row. The revised methodology and the resulting output statistics are presented in the sections below.

2 Overall Match Statistics

Table 1 shows the geocoding match statistics across the entire U.S. dataset.

Table 1: Whole U.S. Geocoding Overall Summary

Metric	Value
Total Properties	1,051,219
Matched	970,482
Tied (Multiple Candidates)	6,697
Matched or Tied	977,179
Strict Match Rate	92.32%
Matched-or-Tied Rate	92.96%

Table 2 shows the breakdown by match type.

Table 2: Whole U.S. Match Type Counts

match_type	rows	pct
One-to-One Matched	970,482	92.32
One-to-One Did Not Match	73,355	6.98
Missing	685	0.07
One-to-Many (Tie)	6,697	0.64

3 Whole U.S. Census Tract Resolution

3.1 Tie Resolution Logic

Let A denote the set of One-to-Many (tie) rows. For each row $i \in A$, let

- C_i = the set of candidate matches returned by the geocoder for row i,
- z_i = the input ZIP code of row i,
- $Z(c)$ = the ZIP code of candidate c,
- $T(c)$ = the census tract of candidate c.

Define the ZIP-matching subset of candidates

$$C_i' \;=\; \{\, c \in C_i \;:\; Z(c) = z_i \,\} \;\subseteq\; C_i.$$

The resolved census tract for row i is then

$$\widehat{T}(i) \;=\; \begin{cases} t, & \text{if } C_i' \neq \emptyset \text{ and } T(c) = t \;\; \forall\, c \in C_i' \quad \text{(ZIP-matching ties)}, \\ t, & \text{else if } T(c) = t \;\; \forall\, c \in C_i \quad \text{(all ties unanimous, fallback)}, \\ \text{ambiguous}, & \text{otherwise.} \end{cases}$$

In words: among the tied candidates for row i, first restrict to those whose ZIP matches the input ZIP (C_i'); if that subset is non-empty and unanimous on a single tract t, output t. Otherwise, fall back to the full candidate set C_i: if *all* tied candidates agree on a single tract, output it. If neither condition holds, mark the row as ambiguous.

Table 3 shows the Whole U.S. dataset (1,051,219 rows) collapsed from the geocoder match output to a single census tract per row. The resolution process was first validated on a San Francisco subset before being applied to the full U.S. dataset; see Appendix 4.1 for those results.

Table 3: Whole U.S. Census Tract Resolution Summary

Resolution Method	Rows	Pct
One-to-One Primary	1,043,837	99.30%
Tie Resolved via ZIP-Matching Ties	3,746	0.36%
Tie Resolved via All Ties Unanimous	314	0.03%
Tract Ambiguous	2,637	0.25%
Tract Missing	685	0.07%
Resolved to Single Tract	**1,047,897**	**99.68%**

Table 4 shows the breakdown for the 6,697 One-to-Many (tie) rows specifically.

Table 4: Whole U.S. Tie Row Tract Resolution

Tie Resolution Outcome	Rows	Pct of Tie Rows	Pct of Total Properties
ZIP-matching ties share the same census tract (resolved)	3,746	55.94%	0.36%
All ties share the same census tract, fallback (resolved)	314	4.69%	0.03%
Tie outputs disagree on census tract (ambiguous)	2,637	39.38%	0.25%
Total tie rows	**6,697**	**100.00%**	**0.64%**

3.2 State-Level Tract Resolution

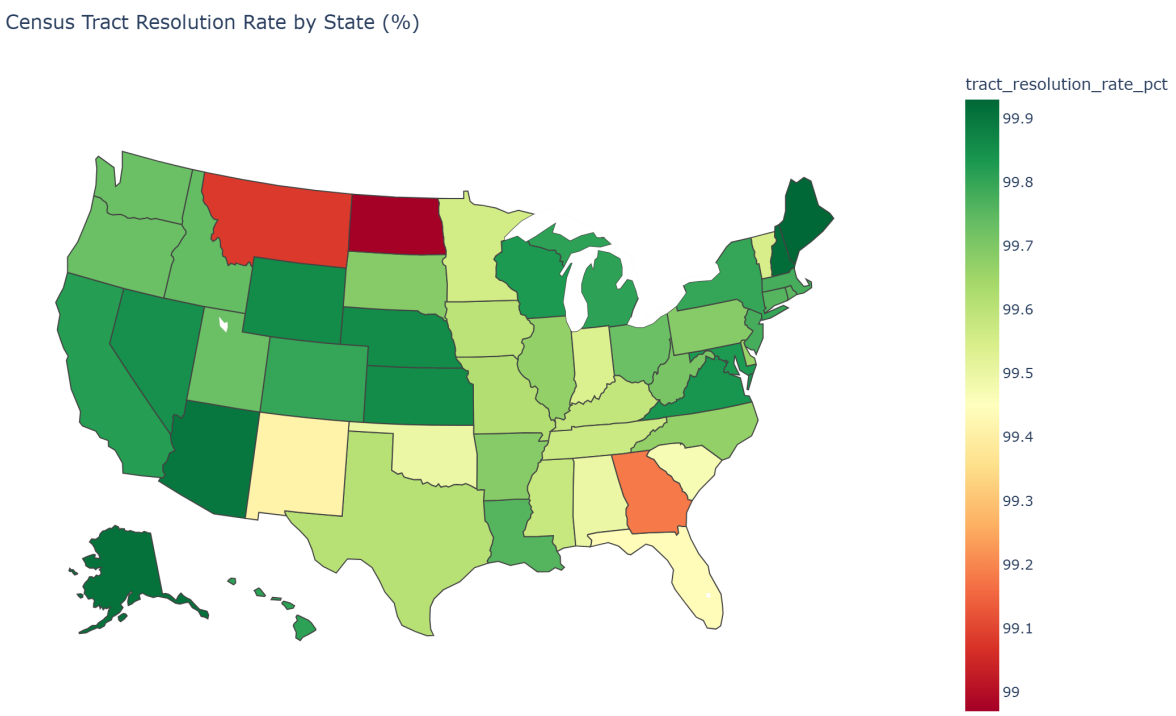

Figure 1: Census Tract Resolution Rate by State

Figure 1 renders the resolution rate for the 50 states + DC. Even the worst-performing states are above 99%, and the variation across the country is small in absolute terms (about 0.75 percentage points between the best and worst).

Table 5 shows the ten largest states by row count and their resolution rates. The `Resolved` column counts rows that the tract matcher successfully collapsed to a single census tract; the `Tract Resolution Rate` column expresses that as a percentage of total rows in each state.

Table 5: Top 10 States by Row Count

State	Total Rows	Resolved	Ambiguous	Missing	Strict Match Rate	Tract Resolution Rate
CA	165,793	165,498	271	24	95.18%	99.82%
TX	88,171	87,826	268	77	87.77%	99.61%

State	Total Rows	Resolved	Ambiguous	Missing	Strict Match Rate	Tract Resolution Rate
FL	79,864	79,414	320	130	91.74%	99.44%
NY	62,311	62,184	99	28	90.15%	99.80%
IL	39,768	39,635	102	31	92.94%	99.67%
PA	37,662	37,547	78	37	90.33%	99.69%
OH	37,188	37,086	81	21	94.44%	99.73%
GA	36,736	36,434	253	49	90.16%	99.18%
NC	34,365	34,251	102	12	93.07%	99.67%
MI	31,221	31,161	49	11	94.47%	99.81%

Notably, while strict geocoder match rates vary considerably across these markets (87.77% in TX vs. 95.18% in CA), tract resolution rates compress to a narrow band (99.18%, 99.82%); the tract matcher's tie-resolution step recovers most of the rows the strict matcher leaves behind.

Table 6 shows every state and territory in the dataset, sorted by strict match rate descending.

Table 6: Match Rate by State (Sorted by Strict Match Rate, Descending)

State	Total	Matched	Tied	Resolved	Strict Match Rate (%)	Tract Resolution Rate (%)
Rhode Island	3,667	3,654	13	3,658	99.65	99.75
New Hampshire	3,782	3,768	14	3,779	99.63	99.92
Massachusetts	21,281	21,199	82	21,234	99.61	99.78
Connecticut	14,131	14,075	56	14,097	99.60	99.76
Vermont	1,101	1,096	5	1,096	99.55	99.55
Maine	2,975	2,957	18	2,973	99.39	99.93
New Jersey	30,168	29,957	211	30,101	99.30	99.78
District of Columbia	3,291	3,187	3	3,285	96.84	99.82
Kansas	7,596	7,296	32	7,585	96.05	99.86
Alaska	1,058	1,016	3	1,057	96.03	99.91
California	165,793	157,804	656	165,498	95.18	99.82
Washington	23,906	22,746	132	23,841	95.15	99.73
Arizona	24,670	23,373	60	24,646	94.74	99.90
Colorado	22,600	21,368	81	22,555	94.55	99.80
Oregon	14,913	14,093	81	14,872	94.50	99.73
Michigan	31,221	29,496	151	31,161	94.47	99.81
Idaho	5,704	5,388	32	5,689	94.46	99.74
Ohio	37,188	35,120	213	37,086	94.44	99.73
Nevada	11,878	11,194	50	11,860	94.24	99.85
Nebraska	4,953	4,666	17	4,946	94.21	99.86
Iowa	5,690	5,356	49	5,667	94.13	99.60
South Dakota	1,602	1,502	10	1,597	93.76	99.69
Tennessee	19,188	17,968	184	19,106	93.64	99.57

State	Total	Matched	Tied	Resolved	Strict Match Rate (%)	Tract Resolution Rate (%)
Maryland	18,351	17,116	77	18,320	93.27	99.83
Virginia	25,455	23,723	79	25,414	93.20	99.84
North Carolina	34,365	31,982	275	34,251	93.07	99.67
Illinois	39,768	36,959	310	39,635	92.94	99.67
Indiana	17,857	16,587	156	17,775	92.89	99.54
Minnesota	13,224	12,239	113	13,166	92.55	99.56
Wisconsin	13,646	12,598	70	13,623	92.32	99.83
Montana	1,959	1,807	24	1,941	92.24	99.08
Utah	10,044	9,256	56	10,017	92.15	99.73
Delaware	2,954	2,714	18	2,944	91.88	99.66
Missouri	17,780	16,327	179	17,713	91.83	99.62
Oklahoma	12,095	11,099	66	12,034	91.77	99.50
Florida	79,864	73,271	809	79,414	91.74	99.44
New Mexico	5,291	4,840	48	5,260	91.48	99.41
South Carolina	19,076	17,431	180	18,974	91.38	99.47
Kentucky	9,212	8,418	78	9,174	91.38	99.59
Arkansas	6,428	5,872	45	6,408	91.35	99.69
Louisiana	10,206	9,275	79	10,182	90.88	99.76
Wyoming	722	656	3	721	90.86	99.86
Pennsylvania	37,662	34,021	213	37,547	90.33	99.69
Georgia	36,736	33,120	479	36,434	90.16	99.18
New York	62,311	56,172	354	62,184	90.15	99.80
Texas	88,171	77,391	556	87,826	87.77	99.61
North Dakota	1,748	1,517	31	1,730	86.78	98.97
Alabama	11,378	9,789	141	11,321	86.03	99.50
Mississippi	4,320	3,655	77	4,302	84.61	99.58
West Virginia	2,724	2,114	17	2,716	77.61	99.71
Hawaii	3,197	2,254	11	3,191	70.50	99.81

4 Appendix

4.1 SF Subset Validation

Before applying the resolution process to the full U.S. dataset, it was tested on a San Francisco subset to validate the approach.

Table 7 shows the SF subset (9,352 rows) collapsed from the geocoder match output to a single census tract per row. One-to-Many (tie) rows resolve when all tie outputs share the same census tract; otherwise they are flagged as ambiguous.

Table 7: SF Census Tract Resolution Summary

Resolution Method	Rows	Pct
One-to-One Primary	8,980	96.02%

Resolution Method	Rows	Pct
Tie Resolved via ZIP-Matching Ties	5	0.05%
Tie Resolved via All Ties Unanimous	0	0.00%
Tract Ambiguous	13	0.14%
Tract Missing	354	3.79%
Resolved to Single Tract	**8,985**	**96.08%**

Table 8 shows the breakdown for the 18 One-to-Many (tie) rows specifically.

Table 8: SF Tie Row Tract Resolution

Tie Resolution Outcome	Rows
ZIP-matching ties share the same census tract (resolved)	5
All ties share the same census tract, fallback (resolved)	0
Tie outputs disagree on census tract (ambiguous)	13
Total tie rows	18

4.2 Code Repository

- https://github.com/WilliamClintC/Geocoding_Office